

SETTLING THE OPTIMAL EXPONENT RELATING SUMSETS AND DIFFERENCE SETS

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ABSTRACT. For a finite nonempty set A in an abelian group, let

$$\sigma(A) = \frac{|A + A|}{|A|}, \quad \delta(A) = \frac{|A - A|}{|A|}.$$

The classical sum–difference inequalities state $\sigma(A)^{1/2} \leq \delta(A) \leq \sigma(A)^2$. The exponent 2 in the second inequality is known to be optimal, whereas it has remained open whether the exponent 1/2 in the first inequality can be improved. We settle this question by constructing an explicit family of finite sets $A_K \subset \mathbb{Z}$ such that

$$\frac{\log \sigma(A_K)}{\log \delta(A_K)} \longrightarrow 2,$$

hence the exponent 1/2 in the first inequality is also optimal.

The construction and its proof were developed with the assistance of Hyra, an AI research agent based on the open-sourced Hy3 model.

1. INTRODUCTION

Let A be a finite nonempty subset of an abelian group. Its doubling and difference constants are

$$\sigma(A) = \frac{|A + A|}{|A|}, \quad \delta(A) = \frac{|A - A|}{|A|}.$$

The classical sum–difference inequalities state

$$\sigma(A)^{1/2} \leq \delta(A) \leq \sigma(A)^2 \tag{1}$$

For a finite set $A \subset \mathbb{Z}$ with $|A| \geq 2$, define

$$C(A) := \frac{\log \sigma(A)}{\log \delta(A)}.$$

The problem is to determine how large $C(A)$ can be, or equivalently to determine the least universal exponent c for which $\sigma(A) \leq \delta(A)^c$. Note that (1) immediately implies an upper bound: for any $A \subset \mathbb{Z}$,

$$C(A) \leq 2 \tag{2}$$

Georgiev, Gómez-Serrano, Tao, and Wagner studied this question using AlphaEvolve [4]. In their study, they used the Freiman–Pigarev value

$$\frac{\log(59/17)}{\log(55/17)} = 1.059793\dots$$

as a baseline, and their AI-assisted search found a construction with $C(A) \approx 1.1219$. Under the present normalization, however, Penman and Wells had previously constructed a sequence (Q_j) of finite integer sets for which

$$C(Q_j) \geq 1.125944$$

for all sufficiently large j ; see [9, 2].

We prove that the classical upper bound (2) is optimal. More precisely, our main result is the following.

Theorem 1.1. *For every positive even integer K , there is an explicitly defined finite set $A_K \subset \mathbb{Z}$ such that*

$$C(A_K) > \frac{2K}{K+3}.$$

Consequently, the least universal exponent is 2.

The sets A_K are described in (12) and (13). The proof combines a base-12 digit gadget, a three-state carry automaton, a symmetric additive basis in a cyclic group, and a Chinese-remainder construction. We include all details below. We also provide a lean-based formal proof at <https://github.com/linhaowei1/sum-diff-proof>.

2. PROOF

2.1. A base-12 digit gadget. Let

$$W = \{0, 1, 2, 4, 5, 9\} \subset \mathbb{Z}.$$

Reducing the corresponding sumset and difference set modulo 12, a direct calculation gives

$$(W + W) \bmod 12 = \mathbb{Z}/12\mathbb{Z}, \quad (W - W) \bmod 12 = (\mathbb{Z}/12\mathbb{Z}) \setminus \{6\}. \quad (3)$$

As an integer difference set,

$$\Delta := W - W = \{-9, -8, -7, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 7, 8, 9\}.$$

For an integer $j \geq 1$, put

$$n_j = 12^j, \quad Y_j = \left\{ \sum_{i=0}^{j-1} w_i 12^i : w_i \in W \right\} \subseteq \{0, 1, \dots, n_j - 1\}.$$

Then

$$|Y_j| = 6^j. \quad (4)$$

Lemma 2.1. *For every $j \geq 1$,*

$$(Y_j + Y_j) \bmod n_j = \mathbb{Z}/n_j\mathbb{Z}.$$

Proof. Since $Y_j = W + 12Y_{j-1}$, the result follows by induction from

$$(W + W) \bmod 12 = \mathbb{Z}/12\mathbb{Z},$$

lifting one base-12 digit at a time. □

We next estimate the modular difference set. Define

$$t_j := |(Y_j - Y_j) \bmod n_j|.$$

Lemma 2.2. *The sequence t_j satisfies*

$$t_0 = 1, \quad t_1 = 11, \quad t_{j+2} = 13t_{j+1} - 16t_j \quad (j \geq 0).$$

In particular, if

$$\lambda = \frac{13 + \sqrt{105}}{2}, \quad \mu = \frac{13 - \sqrt{105}}{2},$$

then

$$t_j = \frac{\sqrt{105} + 9}{2\sqrt{105}}\lambda^j + \frac{\sqrt{105} - 9}{2\sqrt{105}}\mu^j < \lambda^j \quad (j \geq 1). \quad (5)$$

Proof. Every residue modulo 12^j has a unique balanced expansion

$$\sum_{i=0}^{j-1} \varepsilon_i 12^i, \quad \varepsilon_i \in E := \{-6, -5, \dots, 5\}.$$

Such a balanced digit string is represented by a difference of two elements of Y_j if and only if there are carries

$$c_0 = 0, \quad c_i \in \{-1, 0, 1\},$$

such that

$$\delta_i = \varepsilon_i + c_i - 12c_{i+1} \in \Delta \quad (0 \leq i < j). \quad (6)$$

Indeed, summing (6) after multiplication by 12^i shows that the two digit strings differ by $-c_j 12^j$.

After any balanced prefix, the only possible nonempty sets of current carries are

$$X_1 = \{0\}, \quad X_2 = \{-1, 0\}, \quad X_3 = \{0, 1\}.$$

For a current carry-set X and a balanced digit ε , define

$$\Phi_\varepsilon(X) = \{c' \in \{-1, 0, 1\} : \varepsilon + c - 12c' \in \Delta \text{ for some } c \in X\}.$$

A check of the twelve possible values of ε gives the transition counts

	X_1	X_2	X_3
X_1	5	3	3
X_2	4	5	3
X_3	4	4	4

(7)

where rows are current states and columns are next states. Thus, with

$$M = \begin{pmatrix} 5 & 3 & 3 \\ 4 & 5 & 3 \\ 4 & 4 & 4 \end{pmatrix}, \quad e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

we have

$$t_j = e_1^\top M^j \mathbf{1}.$$

A direct multiplication gives

$$M^2 - 13M + 16I_3 = \begin{pmatrix} 0 & 3 & -3 \\ 0 & 0 & 0 \\ 0 & -4 & 4 \end{pmatrix}, \quad (M^2 - 13M + 16I_3)\mathbf{1} = 0.$$

Therefore

$$t_{j+2} - 13t_{j+1} + 16t_j = 0.$$

The initial values are $t_0 = 1$ and $t_1 = 11$. Solving the recurrence gives (5). Both coefficients in (5) are positive and sum to 1, while $0 < \mu < \lambda$, so $t_j < \lambda^j$ for $j \geq 1$. $\square$

The following elementary numerical estimate will be useful.

Lemma 2.3. *We have*

$$\frac{\lambda}{12} < \frac{31}{32}, \quad \left(\frac{31}{32}\right)^{22} < \frac{1}{2}.$$

Proof. Straightforward calculation. $\square$

2.2. A symmetric additive basis in a cyclic group. Fix a positive even integer K and set

$$s = 2^K + 1, \quad Q = s^2, \quad m = \frac{s-1}{2}. \quad (8)$$

Inside $\mathbb{Z}/Q\mathbb{Z}$, define

$$H = \{-m, -m+1, \dots, m\}, \quad V = \{0, s, 2s, \dots, (s-1)s\},$$

and put

$$I = H \cup V, \quad B = I \setminus \{0\}.$$

Since $H \cap V = \{0\}$,

$$\rho := |I| = 2s - 1, \quad |B| = 2s - 2. \quad (9)$$

Both H and V are symmetric, so $I = -I$ in $\mathbb{Z}/Q\mathbb{Z}$.

Lemma 2.4. *The set I is an additive basis of $\mathbb{Z}/Q\mathbb{Z}$:*

$$I + I = \mathbb{Z}/Q\mathbb{Z}.$$

Moreover, every $x \notin I$ lies in both $B + B$ and $B - B$.

Proof. Given $x \in \mathbb{Z}/Q\mathbb{Z}$, let $h \in H$ be the unique balanced representative of x modulo s . Then $v = x - h$ is divisible by s , so $v \in V$, and $x = h + v$. Hence $I + I = \mathbb{Z}/Q\mathbb{Z}$.

If $x \notin I$, then neither h nor v is zero. Thus $h, v \in B$ and $x = h + v \in B + B$. Since $-v \in B$, we also have $x = h - (-v) \in B - B$. $\square$

2.3. The CRT construction. For a fixed positive even integer K , set

$$d = 22(K+2), \quad n = 12^d, \quad q = Qn, \quad Y = Y_d, \quad t = t_d. \quad (10)$$

Because K is even, $2^K \equiv 1 \pmod{3}$; $s = 2^K + 1 \equiv 2 \pmod{3}$. Also s is odd. Therefore $\gcd(Q, n) = 1$, and the Chinese remainder theorem gives

$$\mathbb{Z}/q\mathbb{Z} \cong \mathbb{Z}/Q\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

In this product group define

$$\mathcal{R} = (\{0\} \times \mathbb{Z}/n\mathbb{Z}) \cup (B \times Y). \quad (11)$$

Let $R \subseteq \{0, 1, \dots, q-1\}$ be the set of least nonnegative representatives of the inverse image of $\mathcal{R}$. Equivalently,

$$\begin{aligned} R &= \{x \in \mathbb{Z} : 0 \leq x < q, x \equiv 0 \pmod{Q}\} \\ &\cup \{x \in \mathbb{Z} : 0 \leq x < q, x \bmod Q \in B, x \bmod n \in Y\}. \end{aligned} \quad (12)$$

Finally define the finite integer set A_K (abbreviated to A when K is fixed) by

$$\boxed{A_K = R \cup (R + q) = R + q\{0, 1\}.} \quad (13)$$

The two copies are disjoint, and hence

$$|A| = 2|R| = 2(n + (\rho - 1)6^d). \quad (14)$$

2.4. Exact modular counts.

Lemma 2.5. *Modulo q , the sumset of R is the whole group:*

$$|(R + R) \bmod q| = q = Qn.$$

Proof. Work in $\mathbb{Z}/Q\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$. For outer coordinate 0, the sum of the full zero row with itself is full. For an outer coordinate $b \in B$, the sum of the full zero row and the row $\{b\} \times Y$ is full in the inner coordinate. Finally, if $x \notin I$, Lemma 2.4 gives $x = b_1 + b_2$ with $b_1, b_2 \in B$, and Lemma 2.1 gives $(Y + Y) \bmod n = \mathbb{Z}/n\mathbb{Z}$. Thus every outer coordinate has a full inner fibre. $\square$

Lemma 2.6. *Modulo q ,*

$$|(R - R) \bmod q| = \rho n + (Q - \rho)t.$$

Proof. Again work in the CRT product. The outer coordinates in I have full inner fibres. For outer coordinate 0, use the full zero row minus itself. For $b \in B$, use the row $\{b\} \times Y$ minus the full zero row; symmetry gives the same conclusion for all of $I = -I$.

Now let $x \notin I$. Any difference with outer coordinate x must use two nonzero rows, because using the zero row would give an outer coordinate in I . Hence its inner fibre is contained in $Y - Y$. Conversely, Lemma 2.4 gives $x = b_1 - b_2$ with $b_1, b_2 \in B$, so every element of $Y - Y$ occurs. Thus each of the $Q - \rho$ outer coordinates outside I has an inner fibre of size exactly t . $\square$

2.5. Lifting the modular counts to integers.

Lemma 2.7. *Let $R \subseteq \{0, 1, \dots, q - 1\}$ and let $A = R + q\{0, 1\}$. Then*

$$|A + A| \geq 3|(R + R) \bmod q|, \quad |A - A| \leq 4|(R - R) \bmod q|.$$

Proof. Fix a residue $c \pmod{q}$ occurring in $R + R$. Its ordinary representatives in $R + R$ have the form $c + kq$ with k in a nonempty subset of $\{0, 1\}$. The block sums from $q\{0, 1\} + q\{0, 1\}$ have indices $\{0, 1, 2\}$. Therefore the integers in $A + A$ above the residue c have at least three distinct quotient indices. Summing over residues proves the first inequality.

For differences, the ordinary representatives in $R - R$ above a fixed residue have quotient indices in a nonempty subset of $\{-1, 0\}$, while the block differences have indices $\{-1, 0, 1\}$. The sum of these two index sets has at most four elements. Summing over the modular difference residues proves the second inequality. $\square$

2.6. The final estimate. Define the two small parameters

$$\alpha = \frac{(\rho - 1)6^d}{n}, \quad \beta = \frac{t}{n}.$$

Using (9), (8), and (10), we obtain

$$\alpha = (2s - 2)2^{-d} = 2^{K+1-d} = 2^{-21K-43} < \frac{1}{2}. \quad (15)$$

By Lemmas 2.2 and 2.3,

$$\beta < \left(\frac{\lambda}{12}\right)^d < \left(\frac{31}{32}\right)^{22(K+2)} < 2^{-(K+2)}. \quad (16)$$

Proof of Theorem 1.1. By Lemma 2.5, Lemma 2.7, and (14),

$$\sigma(A) = \frac{|A + A|}{|A|} \geq \frac{3Qn}{2n(1 + \alpha)} = \frac{3Q}{2(1 + \alpha)} > Q = s^2,$$

where the last inequality uses $\alpha < 1/2$.

Likewise, Lemmas 2.6 and 2.7 give

$$\begin{aligned} \delta(A) &\leq \frac{4(\rho n + (Q - \rho)t)}{2n(1 + \alpha)} \\ &= \frac{2(\rho + (Q - \rho)\beta)}{1 + \alpha} < 2\rho + 2Q\beta. \end{aligned}$$

Since $s = 2^K + 1 < 2^{K+1}$, (16) implies

$$Q\beta < \frac{s^2}{2^{K+2}} < \frac{s}{2}.$$

As $\rho = 2s - 1$, it follows that

$$\delta(A) < 2(2s - 1) + s = 5s - 2 < 5s < 8s.$$

Therefore

$$C(A) = \frac{\log \sigma(A)}{\log \delta(A)} > \frac{2 \log s}{\log(8s)} = \frac{2 \log s}{\log s + 3 \log 2}.$$

The function

$$x \mapsto \frac{2x}{x + 3 \log 2}$$

is strictly increasing for $x > 0$, and $\log s > K \log 2$. Hence

$$C(A) > \frac{2K \log 2}{K \log 2 + 3 \log 2} = \frac{2K}{K + 3}.$$

Letting the positive even integer K tend to infinity shows that no universal exponent smaller than 2 can work, while the classical upper bound in (1) shows that the exponent 2 does work. $\square$

Corollary 2.1. *Among finite subsets $A \subset \mathbb{Z}$ with $|A| \geq 2$,*

$$\sup_A C(A) = 2.$$

Proof. The equality characterization of Steps [10] gives $C(A) < 2$ for every admissible A . On the other hand, Theorem 1.1 gives a family with

$$C(A_K) > \frac{2K}{K + 3} \rightarrow 2$$

as the positive even integer K tends to infinity. $\square$

3. USE OF AI TOOLS

AI tools supported the exploratory and optimization stages of this work; the mathematical claims rest on the explicit construction and self-contained proofs above. Following the SimpleTES framework [12], we used Hyra [6], powered by the Hy3 model, to optimize finite-set constructions from scratch, raising the best value in its autonomous finite-search trajectory from approximately 1.14 to 1.21.

SimpleTES evaluates $C(A)$ from an explicitly enumerated Python list, so memory and enumeration costs limit the size of searchable sets. We therefore allowed agents to

TABLE 1. Published and manuscript-internal result comparison.

Date	Method, reference, and system	Value / bound
<i>Published mathematical constructions</i>		
Unknown	Conway eight-point MSTD set [8, 5]	1.0344
1969	Marica [7]	1.0290
1973	Freiman–Pigarev [3]	1.0598
2013	Penman–Wells family [9, 2]	1.1259
<i>Published agent-search results</i>		
2025-11	AlphaEvolve (Gemini 2.0 Pro + Flash) [4]	1.1219
2025-12	LoongFlow (DeepSeek-R1-250528) [11, 1]	1.1035
2026-04	SimpleTES (gpt-oss-120b) [12]	1.1440
2026-04	SimpleTES with post-training (gpt-oss-120b) [12]	1.1449
2026-07	OpenHands (GPT-5.4 medium) [13]	1.0786
2026-07	OpenClaw (GPT-5.4 medium) [13]	1.1099
2026-07	Codex (GPT-5.4 medium) [13]	1.1121
2026-07	EvoMaster (GPT-5.4 medium) [13]	1.1207
<i>Exploratory runs reported only in this manuscript</i>		
2026-07	Claude Fable 5	1.1133
2026-07	Codex (GPT-5.5) with human guidance	1.2851
<i>Theorem proved in this work</i>		
2026-07	Explicit family A_K (this work; Hyra with Hy3)	$\sup C(A) = 2$

propose constructions and supporting arguments in natural language. Using GPT-5.6 Sol to judge the validity of candidate proposals and a search budget of approximately 24 hours, this workflow produced the construction underlying the paper. The LLM-based ranking was used only to guide exploration, not to certify mathematical correctness. We then independently checked the construction and proof, corrected the exposition, and prepared the argument presented here manually.

For context, we also tested Claude Fable 5 and GPT-5.5 with human guidance. Table 1 places the manuscript-internal runs alongside published constructions and agent-search results. We also used GPT 5.6 sol to convert natural-language proofs into verifiable Lean 4 formal proofs.

4. CONCLUSION

We determine the optimal universal exponent relating sumsets and difference sets. For every positive even integer K , our explicit construction satisfies

$$C(A_K) > \frac{2K}{K+3},$$

and hence $C(A_K) \rightarrow 2$ as $K \rightarrow \infty$. Together with the classical inequality $\sigma(A)^{1/2} \leq \delta(A)$, this proves that the least exponent c for which $\sigma(A) \leq \delta(A)^c$ holds for every finite nonempty set $A \subset \mathbb{Z}$ is exactly 2.

The supremum is approached but, by Steps’ characterization of the equality cases, is not attained by any integer set with at least two elements [10].

ACKNOWLEDGEMENTS

We thank Letian Huang and Shuo Zhou from Peking University, Baihe Huang from UC Berkeley, and Honghao Lin from Carnegie Mellon University for helpful discussions. We also thank the Hyra team at Tencent Hunyuan for their work on Hyra.

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